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STLogic: Spatially-aware Temporal Logical Rule for Explainable Forecasting on Battlefield Spatio-Temporal Knowledge Graphs

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Abstract

Spatio-temporal knowledge graphs (STKGs) have been primarily studied for spatially-aware link prediction and completion, yet they remain limited in supporting future link forecasting and explainable reasoning. Embedding-based models lack interpretability for their predictions, while temporal logical rule-based models such as TLogic provide rule-form explanations but fail to incorporate spatial context. This limitation leads to poor candidate discrimination when a single temporal rule generates dozens to hundreds of candidates with identical rule confidence. We address the task of link forecasting on battlefield STKGs and introduce STLogic, an explainable forecasting framework. STLogic first learns temporal logical rules by extracting temporal random walks and replacing entities and timestamps with variables following the TLogic framework. For each rule, STLogic collects historical grounding instances and extracts distance and distance change between entity pairs as spatial features. These features are then modeled using Gaussian-based statistical modeling to construct a rule-specific spatial context model. During inference, candidate predictions are generated through temporal rule grounding, and each candidate's current spatial features are compared against the corresponding rule's spatial distribution to compute a spatial fit score. The spatial fit score is combined with the rule confidence to refine the grounding score, while retaining the existing Noisy-OR aggregation structure for final candidate ranking. Experimental results demonstrate that incorporating rule-specific spatial evidence improves the explainability, reliability, and interpretability of battlefield situation forecasting. Experimental results on a battlefield STKG constructed from real-world observations demonstrate that incorporating rule-specific spatial evidence improves the reliability and interpretability of situation forecasting. Overall, STLogic integrates temporal logical reasoning with rule-specific spatial context, enabling forecasting that is both temporally consistent and spatially plausible.

Keywords: Spatio-temporal knowledge graph, link forecasting, temporal logical rule, explainable reasoning, spatial context modeling